

Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Python, Go, Java, C++, TypeScript, JavaScript, SQL
- **AI & Data:** RAG Architecture, Vector Databases, LangChain, OpenAI API, FAISS, ChromaDB
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, TCP Sockets, Raft Consensus Algorithm
- **Infrastructure & DB:** AWS (EC2), Docker, PostgreSQL, MongoDB, Git, Linux
- **Testing & Validation:** JUnit, Pytest

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 - Present

Personal Project · Python, Go, LangChain, FAISS

- Automated thousands of Software Engineering job postings by architecting an autonomous AI Co-Pilot in Python, leveraging concurrent JobSpy scrapers for anti-bot evasion.
- Delivered uniquely tailored, ATS-optimized PDF resumes by building a zero-hallucination RAG engine using LangChain, OpenAI APIs, and a sandboxed FAISS Vector Store.
- Ensured application state durability and fault tolerance by developing TitanStore, a distributed memory bank in Go, implementing the **Raft Consensus Algorithm** and a custom Write-Ahead Log (WAL).
- Enabled concurrent, real-time status updates to the Go database cluster by designing and implementing a high-throughput TCP client API, supporting seamless scaling to 100+ users.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Optimized deployments and reduced infrastructure costs by migrating a full-stack platform from Azure to AWS (EC2) and implementing Docker container orchestration.
- Increased data throughput by 10x while respecting server constraints by engineering a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting.
- Engineered a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting, increasing data throughput by 10x while respecting server constraints.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).

Gridlock Casino (2D Arcade Engine)

Feb 2026 - Present

Collaborative Project · Java, Swing, Maven, JUnit

- Managed project build automation and dependency resolution using Apache Maven to synchronize a 6-developer team, and validated core engine logic through comprehensive **JUnit testing**.
- Architected a custom 2D grid-based arcade game engine in Java, managing game state, 60-FPS rendering cycles, and concurrent user input within a strict MVC architecture.

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025